

Arithmetic of Binomial Coefficients

1. Prove that the number of odd terms in every row of the Pascal triangle is a power of 2.

2. Let p be a prime number ≥ 5 . Prove that:

$$\binom{2p-1}{p-1} \equiv 1 \pmod{p^3}.$$

3. (Generalization of the previous problem.) Let p be a prime number ≥ 5 . Prove that, for every non-negative integers m, n , we have:

$$\binom{mp}{np} \equiv \binom{m}{n} \pmod{p^3}.$$

4. (IMO2007SL.) Prove that, for every positive integer n ,

$$\binom{2^{n+1}}{2^n} \equiv \binom{2^n}{2^{n-1}} \pmod{2^{3n}},$$

but

$$\binom{2^{n+1}}{2^n} \not\equiv \binom{2^n}{2^{n-1}} \pmod{2^{3n+1}}.$$

5. Prove that for every positive integers n ,

$$\binom{3n}{n} \not\equiv 2 \pmod{4}.$$

6. (China 2015.) Find all non-negative integers k with the following property: there are infinitely many positive integers n such that

$$n + k \nmid \binom{2n}{n}$$

7. (Vietnam TST 2010.) Let S_n denote the sum of square of the coefficients of $(1+x)^n$ (after expansion). Prove that $S_{2n} + 1$ is not divisible by 3.

8. (Vietnam TST 2025.) Find all positive integers k such that there exist infinitely many positive integers n such that $\binom{(2025+k)n}{2025n}$ is not divisible by $kn + 1$.